Chapter 8. Correlation and Regression Analysis

Correlation, simple and multiple regression

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8.1 Introduction

Motivation. Data often come in pairs — IQ and grades, smoking and lifespan, process temperature and product strength — and we want to know whether, and how, the two variables are related. Scatter plots give a first look; correlation and regression analysis make the inference precise.

Correlation analysis studies whether two variables are (linearly) related and how strongly, usually through inference on the population correlation coefficient based on the sample correlation coefficient. **Regression analysis** models one variable as a function of another so that its value can be predicted. When the function is a straight line we speak of **simple linear regression**; when several explanatory variables jointly affect the response, of **multiple regression**.

The terms go back to Francis Galton, who studied the inheritance of sweet-pea seed size and of heights of parents and children; "regression" comes from his phrase *regression toward the mean*.

8.2 Correlation analysis

Motivation. The population correlation ρ measures the strength of a linear relation; its sample analogue r estimates it, and under bivariate normality a t-test decides whether $\rho \neq 0$.

For two characteristics X, Y of the units of a population, the **population correlation coefficient** $\rho = \operatorname{Corr}(X, Y)$ satisfies $-1 \le \rho \le 1$; values near $+1$ (-1) indicate a strong linear relation with positive (negative) slope, values near 0 almost no linear relation. From a random sample $(x_1, y_1), \dots, (x_n, y_n)$, ρ is estimated by the **sample correlation coefficient**

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\,\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}.$$

Also $-1 \le r \le 1$, and $r \to \pm 1$ as the points cluster around a line. A value $r \approx 0$ means a *weak linear* relation, not the absence of any relation: a strongly curved (e.g. parabolic) pattern can have $r = 0$.

Illustration 8-1: cycling distance and calories

Twenty randomly chosen adult men cycled for 10 minutes; distance x (km) and calories burned y (kcal) were recorded.

x	3.4	3.0	4.3	4.5	3.8	4.3	3.2	3.4	3.1	3.1
y	96	84	112	123	111	104	95	105	91	79
x	3.2	3.5	3.2	3.2	3.1	3.4	3.2	3.1	3.9	4.6
y	98	103	96	82	97	87	83	86	100	126

The scatter plot shows an increasing linear trend, and

$$r = \frac{110.88}{\sqrt{5.10}\,\sqrt{3258.55}} = 0.860.$$

Inference on ρ assumes the population is **bivariate normal**, i.e. (X, Y) has density

$$f(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\Big\{-\frac{1}{2(1-\rho^2)}\Big[\Big(\tfrac{x-\mu_X}{\sigma_X}\Big)^2 - 2\rho\,\tfrac{(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y} + \Big(\tfrac{y-\mu_Y}{\sigma_Y}\Big)^2\Big]\Big\},$$

whose marginals are $N(\mu_X, \sigma_X^2)$ and $N(\mu_Y, \sigma_Y^2)$.

Definition (Test for correlation (bivariate normal population))

For $H_0 : \rho = 0$ use $T = \sqrt{n-2}\,\dfrac{r}{\sqrt{1-r^2}}$, which is $t(n-2)$ under H_0. With observed value t:

H_1	P-value	rejection region
(a) $\rho > 0$	$P(T \geq t)$	$t \geq t_\alpha(n-2)$
(b) $\rho < 0$	$P(T \leq t)$	$t \leq -t_\alpha(n-2)$
(c) $\rho \neq 0$	$P(\lvert T\rvert \geq \lvert t\rvert)$	$\lvert t\rvert \geq t_{\alpha/2}(n-2)$

Why $T \sim t(n-2)$

[Supplement] Regress y on x (§8.3). One checks algebraically that $\sqrt{n-2}\,r/\sqrt{1-r^2} = \hat\beta/(\hat\sigma/\sqrt{S_{(xx)}})$, using $r^2 = \mathrm{SSR}/\mathrm{SST}$ and $1-r^2 = \mathrm{SSE}/\mathrm{SST}$. Under bivariate normality with $\rho = 0$, conditionally on the x_i the y_i follow the simple linear regression model with $\beta = 0$, so by §8.4 this ratio is $t(n-2)$ given the x's; since this conditional law does not depend on the x's, it is also the unconditional law.

Example 8-1 (Is distance correlated with calories?)

In Illustration 8-1 assume bivariate normality and test at level 5% whether the two variables are correlated.

Solution. $H_0 : \rho = 0$ vs. $H_1 : \rho \neq 0$; $n = 20$, $r = 0.860$:

$$t = \sqrt{20-2}\,\frac{0.860}{\sqrt{1-0.860^2}} = 7.150, \qquad |t| \geq t_{0.025}(18) = 2.101.$$

H_0 is rejected; the P-value is 0.000001, strong evidence of correlation.

8.3 The simple linear regression model and its fit

Motivation. To predict a response (e.g. product strength) from an explanatory variable (e.g. injection temperature) we postulate a straight-line model for the mean response and fit it by least squares.

The variable that influences is the **explanatory variable**; the one influenced is the **response variable**. Regression first examines the scatter plot to choose a tentative model.

Illustration 8-2: temperature and hot-drink sales

A coffee shop recorded the daily maximum temperature x (°C) and hot-drink sales y (10,000 won) on 12 random days.

x	27.1	9.1	19.2	29.1	22.0	34.9
y	46.8	63.7	46.0	46.1	50.3	34.7
x	16.5	7.6	13.1	2.6	23.5	5.6
y	57.8	63.2	52.7	62.9	45.3	69.9

The points scatter around an unknown line $y = \alpha + \beta x$ with negative slope (Figure 1).

We regard y as an observed value of $Y = \alpha + \beta x + e$, where $E(e) = 0$, $\mathrm{Var}(e) = \sigma^2$; equivalently

$E(Y \mid X = x) = \alpha + \beta x$ and $\mathrm{Var}(Y \mid X = x) = \sigma^2$. This is the **simple linear regression model**; the responses $y_1, \ldots, y_n$ at $x_1, \ldots, x_n$ are observed independently.

Definition (Assumptions of the simple linear regression model)

$Y_i = \alpha + \beta x_i + e_i$ $(i = 1, \ldots, n)$, where
(i) linearity: $E(e_i) = 0$, i.e. $E(Y_i \mid X = x_i) = \alpha + \beta x_i$ $(-\infty < \alpha, \beta < \infty)$;
(ii) equal variance: $\mathrm{Var}(e_1) = \cdots = \mathrm{Var}(e_n) = \sigma^2 > 0$;
(iii) independence: $e_1, \ldots, e_n$ are independent.

The line $y = \alpha + \beta x$ is the **population regression line**, α and β are the **population regression coefficients**, and σ^2 is the **error variance**. Fitting the model means estimating these from data, by the **method of least squares**.

Definition 8-1 (Method of least squares)

In the model $Y_i = \alpha + \beta x_i + e_i$, the estimators of α, β are the values minimizing

$$\sum_{i=1}^{n} \{Y_i - (\alpha + \beta x_i)\}^2.$$

Setting the partial derivatives in α and β to zero gives the **normal equations**

$$\begin{cases} -2\sum(y_i - \hat\alpha - \hat\beta x_i) = 0 \\ -2\sum x_i(y_i - \hat\alpha - \hat\beta x_i) = 0 \end{cases} \quad \text{i.e.} \quad \begin{cases} n\hat\alpha + (\sum x_i)\hat\beta = \sum y_i \\ (\sum x_i)\hat\alpha + (\sum x_i^2)\hat\beta = \sum x_i y_i. \end{cases}$$

The solutions are the **least squares estimators** (LSE), and $\hat y = \hat\alpha + \hat\beta x$ is the **least squares regression line**. Note that $\hat y_i$ estimates the *mean* response $E(Y_i \mid X = x_i)$, not an individual response. With the shorthand

$$S_{(xy)} = \sum(x_i - \bar x)(y_i - \bar y) = \sum x_i y_i - \tfrac{1}{n}(\sum x_i)(\sum y_i), \qquad S_{(xx)} = \sum(x_i - \bar x)^2 = \sum x_i^2 - \tfrac{1}{n}(\sum x_i)^2,$$

and $S_{(yy)}$ defined likewise, the solution is as follows.

Proposition (Least squares estimators and line)

(i) $\hat\alpha = \bar y - \hat\beta \bar x, \quad \hat\beta = S_{(xy)}/S_{(xx)}$.
(ii) The least squares line is $\hat y = \hat\alpha + \hat\beta x = \bar y + \hat\beta(x - \bar x)$.
(iii) The minimum value of the sum of squares is $\mathrm{SSE} = S_{(yy)} - \hat\beta^2 S_{(xx)} = S_{(yy)} - S_{(xy)}^2/S_{(xx)}$.

Proof. [Supplement] Assume $S_{(xx)} > 0$ (not all x_i equal). Put $Q(a, b) = \sum(y_i - a - bx_i)^2$.

Step 1. Write $y_i - a - bx_i = \{(y_i - \bar y) - b(x_i - \bar x)\} + (\bar y - a - b\bar x)$. The cross term vanishes because $\sum(y_i - \bar y) = \sum(x_i - \bar x) = 0$, so

$$Q(a, b) = \sum\{(y_i - \bar y) - b(x_i - \bar x)\}^2 + n(\bar y - a - b\bar x)^2.$$

Step 2. Expanding the first sum and completing the square in b,

$$\sum\{(y_i - \bar y) - b(x_i - \bar x)\}^2 = S_{(yy)} - 2bS_{(xy)} + b^2 S_{(xx)} = S_{(xx)}\Big(b - \frac{S_{(xy)}}{S_{(xx)}}\Big)^2 + S_{(yy)} - \frac{S_{(xy)}^2}{S_{(xx)}}.$$

Step 3. Both squares in Q are ≥ 0, and both vanish exactly when $b = S_{(xy)}/S_{(xx)}$ and $a = \bar y - b\bar x$. Hence this is the unique minimizer, giving (i) and (ii), and the minimum is $S_{(yy)} - S_{(xy)}^2/S_{(xx)}$; since $\hat\beta^2 S_{(xx)} = S_{(xy)}^2/S_{(xx)}$, (iii) follows. The minimizer of course also solves the normal equations. ■

Example 8-2 (Fitting the hot-drink data)

For Illustration 8-2, with temperature as X and sales as Y, find the LSE of α, β and draw the fitted line.

Solution. $\bar{x} = 210.3/12 = 17.525$, $\bar{y} = 639.4/12 = 53.283$,

$$S_{(xy)} = 10134.21 - \tfrac{1}{12} \times 210.3 \times 639.4 = -1071.275,$$
$$S_{(xx)} = 4826.67 - \tfrac{1}{12} \times 210.3^2 = 1141.162.$$

So $\hat{\beta} = S_{(xy)}/S_{(xx)} = -0.939$, $\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x} = 69.739$, and $\hat{y} = 69.739 - 0.939x$ (Figure 1).

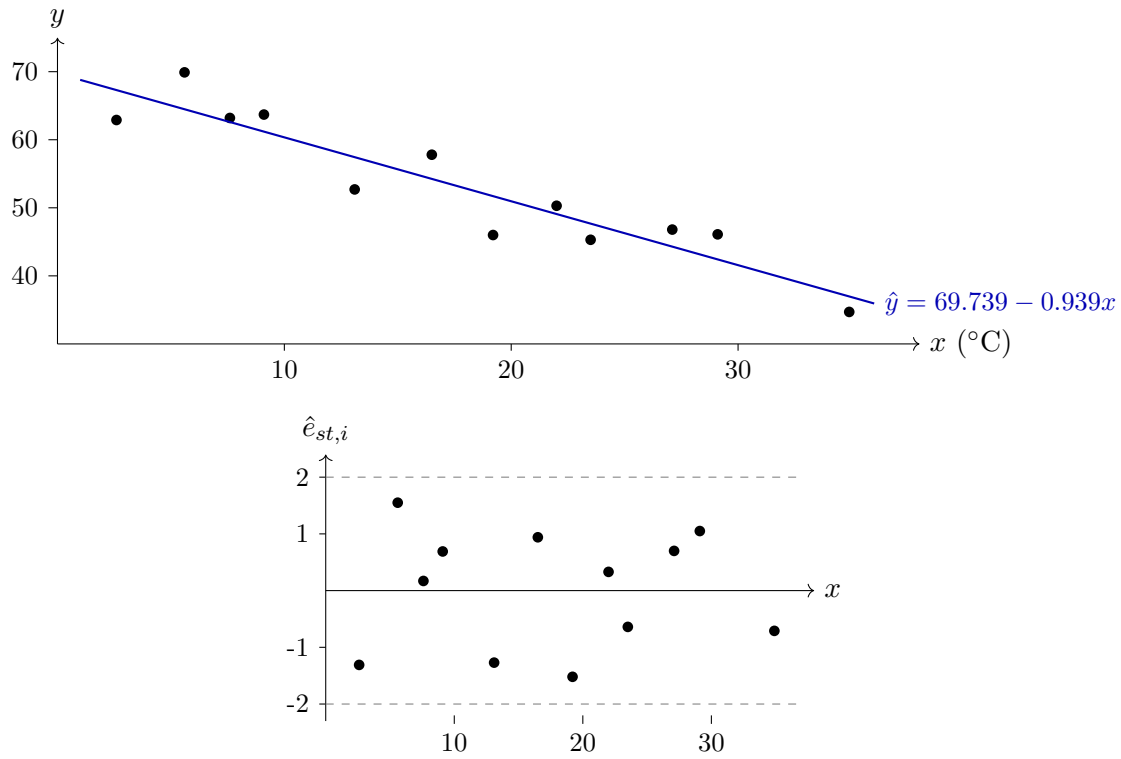


Figure 1: Top: the hot-drink data with the least squares line (Example 8-2). Bottom: studentized residuals against temperature (§8.5); they are symmetric about 0, show no pattern, and lie within ±2.

8.3.1 Residuals and the estimate of σ^2

The **residuals** are $\hat{e}_i = y_i - \hat{y}_i$; small residuals mean a good fit. They behave like observed errors e_i, so their size reflects σ^2. The **residual** (or **error**) **sum of squares** is SSE $= \sum(y_i - \hat{y}_i)^2$, and σ^2 is estimated by

$$\hat{\sigma}^2 = \frac{\text{SSE}}{n-2} = \frac{1}{n-2}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2.$$

The divisor $n - 2$ is the degrees of freedom of SSE, and $\hat{\sigma}^2$ is called the **mean squared error**, MSE.

Definition (Residual sum of squares and the estimate of σ^2)

(i) $\text{SSE} = \sum(y_i - \hat{y}_i)^2 = \sum\{y_i - \bar{y} - \hat{\beta}(x_i - \bar{x})\}^2 = S_{(yy)} - \hat{\beta}^2 S_{(xx)} = S_{(yy)} - S^2_{(xy)}/S_{(xx)}$.

(ii) $\hat{\sigma}^2 = \text{MSE} = \text{SSE}/(n-2)$.

Proposition (MSE is unbiased)

Under the model assumptions, $E(\text{SSE}) = (n-2)\sigma^2$, so $E(\text{MSE}) = \sigma^2$.

Proof. [Supplement] Treat the x_i as fixed.

Step 1. $\sum(Y_i - \bar{Y})^2 = \sum Y_i^2 - n\bar{Y}^2$ with $E(Y_i^2) = \sigma^2 + \mu_i^2$, $\mu_i = \alpha + \beta x_i$, and $E(\bar{Y}^2) = \sigma^2/n + \bar{\mu}^2$. Hence $E\,S_{(YY)} = (n-1)\sigma^2 + \sum(\mu_i - \bar{\mu})^2 = (n-1)\sigma^2 + \beta^2 S_{(xx)}$.

Step 2. By the next proposition, $E(\hat{\beta}) = \beta$ and $\text{Var}(\hat{\beta}) = \sigma^2/S_{(xx)}$, so $E(\hat{\beta}^2)S_{(xx)} = \sigma^2 + \beta^2 S_{(xx)}$.

Step 3. Subtracting, $E(\text{SSE}) = E\,S_{(YY)} - E(\hat{\beta}^2)S_{(xx)} = (n-2)\sigma^2$. ■

Example 8-3 (Estimating σ^2)

For the hot-drink data, estimate the error variance.
Solution. $S_{(xy)} = -1071.275$, $S_{(xx)} = 1141.162$, $S_{(yy)} = 1160.837$, so

$$\text{SSE} = 1160.837 - \frac{(-1071.275)^2}{1141.162} = 155.169, \qquad \hat{\sigma}^2 = \text{MSE} = \frac{155.169}{10} = 15.517.$$

8.3.2 Decomposition of the total sum of squares

How much of the variation of y does the fitted line explain? The deviation of y_i from $\bar{y}$ splits as

$$y_i - \bar{y} = (y_i - \hat{y}_i) + (\hat{y}_i - \bar{y}), \tag{8.1}$$

where $y_i - \hat{y}_i$ is the residual (due to error) and $\hat{y}_i - \bar{y}$ is the part explained by the regression line. $y_i - \bar{y}$ is the **total deviation** and (8.1) its **decomposition**. Squaring and summing gives a Pythagoras-type identity. Define the **total sum of squares** $\text{SST} = \sum(y_i - \bar{y})^2$ (d.f. $n-1$) and the **regression sum of squares** $\text{SSR} = \sum(\hat{y}_i - \bar{y})^2$ (d.f. 1).

Proposition (Decomposition of the total sum of squares)

$$\underbrace{\sum_{i=1}^{n}(y_i - \bar{y})^2}_{\text{SST},\ n-1} = \underbrace{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}_{\text{SSE},\ n-2} + \underbrace{\sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2}_{\text{SSR},\ 1}.$$

Moreover $\text{SSR} = \hat{\beta}^2 S_{(xx)} = S_{(xy)}^2/S_{(xx)}$ and $\text{SST} = S_{(yy)}$.

Proof. [Supplement]

Step 1. The normal equations say $\sum \hat{e}_i = 0$ and $\sum x_i \hat{e}_i = 0$, where $\hat{e}_i = y_i - \hat{y}_i$.

Step 2. Since $\hat{y}_i - \bar{y} = \hat{\beta}(x_i - \bar{x})$, the cross term is

$$\sum(y_i - \hat{y}_i)(\hat{y}_i - \bar{y}) = \hat{\beta}\sum \hat{e}_i(x_i - \bar{x}) = \hat{\beta}\Big(\sum x_i\hat{e}_i - \bar{x}\sum \hat{e}_i\Big) = 0.$$

Step 3. Squaring (8.1) and summing therefore gives $\text{SST} = \text{SSE} + \text{SSR}$, and $\text{SSR} = \hat{\beta}^2\sum(x_i - \bar{x})^2 = \hat{\beta}^2 S_{(xx)} = S_{(xy)}^2/S_{(xx)}$. The degrees of freedom $n-1$ and $n-2$ reflect the one and two estimated parameters ($\bar{y}$; $\hat{\alpha}, \hat{\beta}$), and $1 = (n-1) - (n-2)$. ■

The larger the share of SSR in SST, the better the model explains the data. This share is the **coefficient of determination**

$$r^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}} = \frac{S_{(xy)}^2}{S_{(xx)}S_{(yy)}} = \Big(\frac{S_{(xy)}}{\sqrt{S_{(xx)}}\sqrt{S_{(yy)}}}\Big)^2,$$

which is exactly the square of the sample correlation coefficient. Values near 1 mean the points cluster tightly around the line.

Example 8-4 (Coefficient of determination)

Compute and interpret r^2 for the hot-drink data.
Solution. $r^2 = (-1071.275)^2/(1141.162 \times 1160.837) = 0.866$: the line explains 86.6% of the variation in sales, and together with the scatter plot the model describes the data well.

Statistics alone can mislead (Anscombe)

The two data sets below have nearly identical statistics: the same fitted line $\hat{y} = 3 + 0.5x$, SST $= 41.29$, SSR $= 27.51$, SSE $= 13.78$, $\hat{\sigma}^2 = 1.53$, $r^2 = 0.67$. Yet A is roughly linear with scatter, while B lies on a smooth curve. Always look at the scatter plot and the residuals (§8.5).

A	x	10	8	13	9	11	14	6	4	12	7	5
	y	8.04	6.95	7.58	8.81	8.33	9.96	7.24	4.26	10.84	4.82	5.68
B	x	10	8	13	9	11	14	6	4	12	7	5
	y	9.14	8.14	8.74	8.77	9.26	8.10	6.13	3.10	9.13	7.26	4.74

8.4 Inference in simple linear regression

Motivation. Adding the assumption that the errors are normal turns the least squares fit into a basis for confidence intervals and tests about the slope, the intercept and the mean response.

From now on assume, in addition,

$$Y_i = \alpha + \beta x_i + e_i, \qquad e_i \overset{\text{i.i.d.}}{\sim} N(0, \sigma^2) \qquad (i = 1, \ldots, n).$$

8.4.1 Significance of the regression line

The line is useful iff $\beta \neq 0$, so we test $H_0 : \beta = 0$ vs. $H_1 : \beta \neq 0$. A significant line makes SSR large relative to SSE. Under H_0,

$$F = \frac{\text{SSR}/1}{\text{SSE}/(n-2)} \sim F(1, n-2),$$

and with observed value f the P-value is $P(F \geq f)$; the level-α critical region is $f \geq F_\alpha(1, n-2)$. The test is summarized in an **analysis of variance table**; a **mean square** is a sum of squares divided by its degrees of freedom.

Definition (ANOVA table for the significance of the regression line)

Source	SS	d.f.	Mean square	F	P-value
Regression	SSR	1	$\text{MSR} = \text{SSR}/1$	$f = \text{MSR}/\text{MSE}$	$P(F \geq f)$
Residual	SSE	$n-2$	$\text{MSE} = \text{SSE}/(n-2)$		
Total	SST	$n-1$			

Example 8-5 (Is the regression significant?)

Test the significance of the regression line for the hot-drink data.
Solution. SST $= S_{(yy)} = 1160.837$, SSR $= (-1071.275)^2/1141.162 = 1005.668$, SSE $= 155.169$, so MSR $= 1005.668$, MSE $= 15.517$ and $f = 64.811$. Since $P(F(1,10) \geq 64.811) = 0.00001 < 0.001$, the line is highly significant.

Source	SS	d.f.	MS	F	P
Regression	1005.668	1	1005.668	64.811	0.00001
Residual	155.169	10	15.517		
Total	1160.837	11			

8.4.2 Inference on the slope β

Proposition (Distribution of $\hat{\beta}$)

$E(\hat{\beta}) = \beta$, $\text{Var}(\hat{\beta}) = \sigma^2/S_{(xx)}$, $\hat{\beta}$ is normal, and

$$\frac{\hat{\beta} - \beta}{\hat{\sigma}/\sqrt{S_{(xx)}}} \sim t(n-2).$$

Proof. [Supplement]

Step 1. Let $c_i = (x_i - \bar{x})/S_{(xx)}$. Since $\sum(x_i - \bar{x})\bar{Y} = 0$, $\hat{\beta} = \sum c_i Y_i$, a linear combination of independent normal variables, hence normal. Note $\sum c_i = 0$, $\sum c_i x_i = \sum c_i(x_i - \bar{x}) = 1$, $\sum c_i^2 = 1/S_{(xx)}$.

Step 2. $E(\hat{\beta}) = \sum c_i(\alpha + \beta x_i) = \alpha \sum c_i + \beta \sum c_i x_i = \beta$.

Step 3. By independence, $\text{Var}(\hat{\beta}) = \sum c_i^2 \sigma^2 = \sigma^2/S_{(xx)}$. Thus $Z = (\hat{\beta} - \beta)/(\sigma/\sqrt{S_{(xx)}}) \sim N(0,1)$.

Step 4. It is a standard fact of normal linear models (proved via Cochran's theorem) that $\text{SSE}/\sigma^2 \sim \chi^2(n-2)$ and that SSE is independent of $(\hat{\alpha}, \hat{\beta})$. Then $(\hat{\beta}-\beta)/(\hat{\sigma}/\sqrt{S_{(xx)}}) = Z/\sqrt{(\text{SSE}/\sigma^2)/(n-2)}$ is $t(n-2)$ by definition of the t distribution. (Independence is at least easy to see at the level of covariances: $\text{Cov}(\hat{e}_j, \hat{\beta}) = \sigma^2(c_j - \frac{1}{n}\sum_i c_i - (x_j - \bar{x})\sum_i c_i^2) = 0$, and jointly normal uncorrelated variables are independent.) ∎

Definition (Inference on β)

(i) $100(1-\alpha)\%$ CI: $\hat{\beta} \pm t_{\alpha/2}(n-2)\,\hat{\sigma}/\sqrt{S_{(xx)}}$.

(ii) For $H_0 : \beta = \beta_0$: $T = (\hat{\beta} - \beta_0)/(\hat{\sigma}/\sqrt{S_{(xx)}}) \sim t(n-2)$. With observed t:

(a) $H_1 : \beta > \beta_0$: $P = P(T \geq t)$, reject if $t \geq t_\alpha(n-2)$;

(b) $H_1 : \beta < \beta_0$: $P = P(T \le t)$, reject if $t \le -t_\alpha(n-2)$;
(c) $H_1 : \beta \ne \beta_0$: $P = P(|T| \ge |t|)$, reject if $|t| \ge t_{\alpha/2}(n-2)$.

Fact ($F = T^2$)

For $H_0 : \beta = 0$, the ANOVA statistic equals the square of the t-statistic, so the two-sided t-test and the F-test are the same test.

Proof. Using $\mathrm{SSR} = \hat\beta^2 S_{(xx)}$ and $\mathrm{SSE}/(n-2) = \mathrm{MSE} = \hat\sigma^2$,

$$F = \frac{\mathrm{SSR}/1}{\mathrm{SSE}/(n-2)} = \frac{\hat\beta^2 S_{(xx)}}{\mathrm{MSE}} = \frac{\hat\beta^2}{\mathrm{MSE}/S_{(xx)}} = \Big(\frac{\hat\beta - 0}{\hat\sigma/\sqrt{S_{(xx)}}}\Big)^2 = T^2.$$

Since $t_{\alpha/2}(n-2)^2 = F_\alpha(1, n-2)$, the rejection regions coincide. ■

Example 8-6 (Inference on the slope)

For the hot-drink data ($n = 12$, $\hat\beta = -0.939$, $S_{(xx)} = 1141.162$, $\hat\sigma^2 = 15.517$):
(a) test the significance of the line by a t-test at level 1% and relate it to the F-test;
(b) test at level 5% whether a 10°C rise lowers mean sales by more than 90,000 won;
(c) give a 95% CI for β and interpret it.
Solution.
(a) $H_0 : \beta = 0$ vs. $H_1 : \beta \ne 0$: $t = -0.939/\sqrt{15.517/1141.162} = -8.053$, and $|t| \ge t_{0.005}(10) = 3.169$, so the line is significant at 1%. Also $F = T^2$ (here $8.053^2 \approx 64.8$), so the F-test is the same test.
(b) Sales are in units of 10,000 won, so the claim is a drop of more than 0.9 per 1°C: $H_0 : \beta = -0.9$ vs. $H_1 : \beta < -0.9$. $t = (-0.939 + 0.9)/\sqrt{15.517/1141.162} = -0.334 > -t_{0.05}(10) = -1.812$, so H_0 is not rejected; there is no clear evidence of a drop exceeding 90,000 won per 10°C.
(c) $t_{0.025}(10) = 2.228$: $-0.939 \pm 2.228\sqrt{15.517/1141.162} = -0.939 \pm 0.260$, i.e. $(-1.199, -0.679)$. With 95% confidence, each 1°C rise lowers mean hot-drink sales by between 6,790 and 11,990 won.

8.4.3 Inference on the mean response $\alpha + \beta x$

The mean response $E(Y \mid X = x) = \alpha + \beta x$ is estimated by $\hat\alpha + \hat\beta x$.

Proposition (Distribution of $\hat\alpha + \hat\beta x$)

$E(\hat\alpha + \hat\beta x) = \alpha + \beta x$, $\mathrm{Var}(\hat\alpha + \hat\beta x) = \Big(\frac{1}{n} + \frac{(x - \bar x)^2}{S_{(xx)}}\Big)\sigma^2$, it is normal, and

$$\frac{(\hat\alpha + \hat\beta x) - (\alpha + \beta x)}{\hat\sigma\sqrt{n^{-1} + (x - \bar x)^2/S_{(xx)}}} \sim t(n-2).$$

Proof. [Supplement]

Step 1. $\hat\alpha + \hat\beta x = \bar Y + \hat\beta(x - \bar x) = \sum(\frac{1}{n} + (x - \bar x)c_i)Y_i$ is a linear combination of independent normals, hence normal, with mean $(\alpha + \beta\bar x) + \beta(x - \bar x) = \alpha + \beta x$ (using $E\hat\beta = \beta$).

Step 2. $\operatorname{Cov}(\bar{Y}, \hat{\beta}) = \sum_i \frac{1}{n} c_i \sigma^2 = 0$ since $\sum c_i = 0$. Therefore

$$\operatorname{Var}(\hat{\alpha} + \hat{\beta} x) = \operatorname{Var}(\bar{Y}) + (x - \bar{x})^2 \operatorname{Var}(\hat{\beta}) = \frac{\sigma^2}{n} + \frac{(x - \bar{x})^2 \sigma^2}{S_{(xx)}}.$$

Step 3. Standardizing and replacing σ by $\hat{\sigma}$, the $t(n-2)$ law follows exactly as in Step 4 of the previous proof (SSE is independent of $(\bar{Y}, \hat{\beta})$). ■

Definition (Inference on the mean response $E(Y \mid X = x) = \alpha + \beta x$)

(i) $100(1-\alpha)\%$ CI: $(\hat{\alpha} + \hat{\beta} x) \pm t_{\alpha/2}(n-2)\, \hat{\sigma} \sqrt{n^{-1} + (x - \bar{x})^2 / S_{(xx)}}$.

(ii) For $H_0 : \alpha + \beta x = \mu_0$: $T = \dfrac{(\hat{\alpha} + \hat{\beta} x) - \mu_0}{\hat{\sigma} \sqrt{n^{-1} + (x - \bar{x})^2 / S_{(xx)}}} \sim t(n-2)$, with the usual one- or two-sided P-values and critical regions $t \geq t_\alpha(n-2)$, $t \leq -t_\alpha(n-2)$, $|t| \geq t_{\alpha/2}(n-2)$.

Example 8-7 (Mean sales at given temperatures)

For the hot-drink data ($n = 12$, $\bar{x} = 17.525$, $S_{(xx)} = 1141.162$, $\hat{y} = 69.739 - 0.939x$, $\hat{\sigma}^2 = 15.517$):
(a) at $x = 25°$C, test at level 1% whether mean sales exceed 400,000 won;
(b) find 95% CIs for mean sales at $x = 10, 20, 30°$C and plot them.
Solution.
(a) $H_0 : \alpha + 25\beta = 40$ vs. $H_1 : \alpha + 25\beta > 40$; $t = 4.372 \geq t_{0.01}(10) = 2.764$, so H_0 is rejected: mean sales at 25°C clearly exceed 400,000 won.
(b) At $x = 10$:

$$(69.739 - 0.939 \times 10) \pm 2.228 \sqrt{15.517} \sqrt{\tfrac{1}{12} + \tfrac{(10 - 17.525)^2}{1141.162}} = 60.349 \pm 3.200,$$

i.e. (57.149, 63.549). Similarly (48.345, 53.573) at $x = 20$ and (37.455, 45.683) at $x = 30$. Joining the upper and lower limits over all x gives a band, narrowest at $x = \bar{x}$ (a "bow-tie" shape).

The CI for the mean response is shorter when x is close to $\bar{x}$ and when $S_{(xx)}$ is large; so it pays to observe the response at widely spread x values around the x of interest.

8.4.4 Inference on the intercept α

Taking $x = 0$ above:

Definition (Inference on α)

(i) $100(1-\alpha)\%$ CI: $\hat{\alpha} \pm t_{\alpha/2}(n-2)\, \hat{\sigma} \sqrt{n^{-1} + \bar{x}^2 / S_{(xx)}}$.

(ii) For $H_0 : \alpha = \alpha_0$: $T = (\hat{\alpha} - \alpha_0) / (\hat{\sigma} \sqrt{n^{-1} + \bar{x}^2 / S_{(xx)}}) \sim t(n-2)$, with the usual P-values and critical regions.

Example 8-8 (CI for the intercept)

Find a 95% CI for α in the hot-drink model.
Solution. $\hat{\alpha} \pm t_{0.025}(10) \hat{\sigma} \sqrt{1/12 + 17.525^2 / 1141.162} = 69.739 \pm 5.210$, i.e. (64.529, 74.949).

8.5 Residual analysis in simple linear regression

Motivation. All of §8.4 is valid only if the model assumptions hold. Residuals are the observable proxies for the errors, and plotting them is the standard way to check the assumptions.

The full model is $Y_i = \alpha + \beta x_i + e_i$ with (i) linearity $E(e_i) = 0$;

(ii) equal variance $\mathrm{Var}(e_i) = \sigma^2$;

(iii) independence of $e_1, \ldots, e_n$;

(iv) normality $e_i \sim N(0, \sigma^2)$. If they hold, $e_i/\mathrm{SD}(e_i) = (Y_i - (\alpha + \beta x_i))/\sigma$ are independent $N(0,1)$. Replacing e_i by $\hat{e}_i$ and dividing by an estimate of its standard deviation gives the **studentized residuals**

$$\hat{e}_{st,i} = \frac{\hat{e}_i}{\widehat{\mathrm{SD}}(\hat{e}_i)} = \frac{y_i - \hat{y}_i}{\widehat{\mathrm{SD}}(Y_i - \hat{Y}_i)} \qquad (i = 1, \ldots, n),$$

which should look like n independent standard normal observations. (In simple regression $\mathrm{Var}(\hat{e}_i) = \sigma^2\{1 - \frac{1}{n} - (x_i - \bar{x})^2/S_{(xx)}\}$, estimated by plugging in $\hat{\sigma}^2$.) Checking assumptions this way is **residual analysis**, usually via a **residual plot** of the points $(x_i, \hat{e}_{st,i})$.

Definition (Residual analysis)

Compute $\hat{e}_{st,i} = \hat{e}_i/\widehat{\mathrm{SD}}(\hat{e}_i)$ and judge whether they look like n independent $N(0,1)$ observations. The model is adequate if the residuals
(i) are roughly symmetric about 0;
(ii) have constant spread across x;
(iii) show no systematic pattern;
(iv) lie about 95% within ± 2.

The residual plot for the hot-drink data (Figure 1, bottom) satisfies (i)–(iv), so the simple linear regression model is appropriate there and the inferences of §8.4 are meaningful.

Recommended workflow

(i) Draw a scatter plot to see the rough linear trend.
(ii) Fit the least squares line and MSE, then check the studentized residuals (symmetric about 0, mostly within ± 2, no pattern).
(iii) If nothing is wrong, carry out CIs and tests.
Typical violations show up as: a curved pattern (linearity fails); a funnel of growing spread (equal variance fails); runs or waves in order (independence fails); heavy skew or outliers (normality fails). Then a different regression method is needed.

Example 8-9 (Electricity bills and temperature)

Monthly mean temperature x (°C) and the monthly electricity bill y (1,000 won) of an apartment household:

x	y	x	y	x	y	x	y	x	y
−2.2	14.0	0.8	12.2	8.2	12.9	15.2	11.9	21.8	10.3
−1.9	13.1	1.8	13.0	9.2	11.3	16.3	10.9	22.3	9.1
−1.7	14.4	3.9	11.3	14.2	10.8	21.1	9.6	23.6	7.5
−1.3	13.1	7.0	10.5	14.5	10.0	21.1	8.1	23.6	10.5
−0.7	14.8	8.0	9.7	14.9	9.9	21.5	9.2	24.8	10.0

(a) Check by residual analysis whether simple linear regression is appropriate.

(b) Test the significance of the regression line at level 5%.
(c) Give a 95% CI for the mean bill when the mean temperature is 10°C.
Solution. The scatter plot shows a rough decreasing linear relation.
(a) $n = 25$, $\bar{x} = 11.44$, $\bar{y} = 11.124$, $S_{(xx)} = 2210.44$, $S_{(yy)} = 88.8856$, $S_{(xy)} = -374.9945$. So $\hat{\beta} = -0.17$, $\hat{\alpha} = 13.06$, $\hat{y} = 13.06 - 0.17x$, and $\hat{\sigma}^2 = \{S_{(yy)} - S^2_{(xy)}/S_{(xx)}\}/(n-2) = 1.0987$. The studentized residuals are symmetric about 0, have constant spread, show no pattern and all lie within ±2: the model is adequate. Also $r^2 = S^2_{(xy)}/(S_{(xx)}S_{(yy)}) = 0.7157$, so about 72% of the variation is explained.
(b) SST = 88.8856, SSR $= S^2_{(xy)}/S_{(xx)} = 63.6165$, SSE = 25.2691 (d.f. 24, 1, 23); $f = (\text{SSR}/1)/(\text{SSE}/23) = 57.902 \geq F_{0.05}(1, 23) = 4.28$, $P = 0.0000001$: the line is significant.

Source	SS	d.f.	MS	F	P
Regression	63.6165	1	63.6165	57.902	0.0000001
Residual	25.2691	23	1.0987		
Total	88.8856	24			

(c) $\hat{\alpha} + 10\hat{\beta} = 13.06 - 1.7 = 11.36$ with standard error $\hat{\sigma}\sqrt{1/n + (10-\bar{x})^2/S_{(xx)}} = 0.2121$; $t_{0.025}(23) = 2.069$ gives 11.36 ± 0.44, i.e. (10.92, 11.80).

8.6 Multiple regression analysis

Motivation. A response is often affected by several explanatory variables at once; using them all in one model usually explains and predicts better than any single-variable regression.

A regression model with two or more explanatory variables is a **multiple regression model**. With p explanatory variables, the **multiple linear regression model** is

$$Y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \cdots + \beta_p x_{pi} + e_i, \qquad e_i \overset{\text{i.i.d.}}{\sim} N(0, \sigma^2) \quad (i = 1, \ldots, n),$$

with unknown real coefficients $\beta_0, \ldots, \beta_p$ and error variance $\sigma^2 > 0$.

8.6.1 Two explanatory variables

For $Y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + e_i$, least squares minimizes $\sum\{Y_i - (\beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i})\}^2$. Setting the three partial derivatives to zero gives the normal equations

$$\begin{aligned} -2\sum(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{1i} - \hat{\beta}_2 x_{2i}) &= 0, \\ -2\sum x_{1i}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{1i} - \hat{\beta}_2 x_{2i}) &= 0, \\ -2\sum x_{2i}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{1i} - \hat{\beta}_2 x_{2i}) &= 0. \end{aligned}$$

The first gives $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x}_1 - \hat{\beta}_2\bar{x}_2$; substituting into the other two yields

$$S_{(11)}\hat{\beta}_1 + S_{(12)}\hat{\beta}_2 = S_{(1y)}, \qquad S_{(12)}\hat{\beta}_1 + S_{(22)}\hat{\beta}_2 = S_{(2y)},$$

where $S_{(11)} = S_{(x_1x_1)}$, $S_{(22)} = S_{(x_2x_2)}$, $S_{(12)} = S_{(x_1x_2)}$, $S_{(1y)} = S_{(x_1y)}$, $S_{(2y)} = S_{(x_2y)}$.

Definition (LSE in multiple linear regression with two explanatory variables)

$\hat\beta_0, \hat\beta_1, \hat\beta_2$ solve

$$\begin{cases} S_{(11)}\hat\beta_1 + S_{(12)}\hat\beta_2 = S_{(1y)} \\ S_{(12)}\hat\beta_1 + S_{(22)}\hat\beta_2 = S_{(2y)} \\ \hat\beta_0 = \bar y - \hat\beta_1\bar x_1 - \hat\beta_2\bar x_2 \end{cases}$$

and the regression plane $E(Y \mid x_1, x_2) = \beta_0+\beta_1x_1+\beta_2x_2$ is estimated by $\hat y = \hat\beta_0+\hat\beta_1x_1+\hat\beta_2x_2$.

Derivation of the reduced equations. [Supplement] Substitute $\hat\beta_0 = \bar y - \hat\beta_1\bar x_1 - \hat\beta_2\bar x_2$ into the second normal equation: $\sum x_{1i}\{(y_i-\bar y) - \hat\beta_1(x_{1i}-\bar x_1) - \hat\beta_2(x_{2i}-\bar x_2)\} = 0$. Since each bracketed deviation sums to zero over i, x_{1i} may be replaced by $x_{1i}-\bar x_1$, giving $S_{(1y)} - \hat\beta_1 S_{(11)} - \hat\beta_2 S_{(12)} = 0$. The third equation gives the other one in the same way. (As in §8.3, the sum of squares is a convex quadratic, so when $S_{(11)}S_{(22)} - S_{(12)}^2 > 0$ this unique stationary point is the minimum.) ■

Example 8-10 (Standing long jump, age and body fat)

For 10 adult men at a fitness centre, Y = standing long jump (cm), X_1 = age, X_2 = body-fat percentage. Fit the multiple linear regression model.

age	43	64	29	22	31	20	20	26	20	19
body fat	22.7	31.9	12.3	18.4	16.4	26.9	16.0	17.5	22.5	14.5
jump	206	120	250	224	219	195	260	190	226	222

Solution. $n = 10$, $\bar x_1 = 29.4$, $\bar x_2 = 19.91$, $\bar y = 211.2$; $S_{(11)} = 1824.4$, $S_{(22)} = 329.989$, $S_{(12)} = 472.56$, $S_{(1y)} = -3809.8$, $S_{(2y)} = -1723.02$. The normal equations

$$\begin{aligned} 1824.4\,\hat\beta_1 + 472.56\,\hat\beta_2 &= -3809.8, \\ 472.56\,\hat\beta_1 + 329.989\,\hat\beta_2 &= -1723.02, \\ \hat\beta_0 &= 211.2 - 29.4\hat\beta_1 - 19.91\hat\beta_2 \end{aligned}$$

give $\hat\beta_0 = 316.198$, $\hat\beta_1 = -1.1696$, $\hat\beta_2 = -3.5465$, so $\hat y = 316.198 - 1.1696x_1 - 3.5465x_2$.

In practice multiple regression is computed by software; understanding the model, the procedure and the concepts matters more than the arithmetic.

8.6.2 ANOVA in multiple regression

As before $y_i - \bar y = (y_i - \hat y_i) + (\hat y_i - \bar y)$ and

$$\underbrace{\sum (y_i-\bar y)^2}_{\text{SST}} = \underbrace{\sum (y_i-\hat y_i)^2}_{\text{SSE}} + \underbrace{\sum (\hat y_i-\bar y)^2}_{\text{SSR}},$$

with degrees of freedom $n-1$, $n-p-1$ and p. ([Supplement] The proof is the same as in §8.3: the normal equations say that the residuals satisfy $\sum \hat e_i = 0$ and $\sum x_{ki}\hat e_i = 0$ for each k, so $\sum \hat e_i(\hat y_i - \bar y) = 0$ because $\hat y_i - \bar y$ is a linear combination of $1, x_{1i}, \dots, x_{pi}$.) Define

$$\text{MSE} = \frac{\text{SSE}}{n-p-1}, \qquad \text{MSR} = \frac{\text{SSR}}{p}, \qquad \hat\sigma^2 = \text{MSE}, \qquad R^2 = \frac{\text{SSR}}{\text{SST}}.$$

(R^2 is the multiple-regression analogue of r^2.) To test the regression function,

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_p = 0 \qquad \text{vs.} \qquad H_1 : \text{not all } \beta_k \text{ are } 0,$$

use $F = \text{MSR}/\text{MSE}$, which is $F(p, n-p-1)$ under H_0; reject at level α if $f \geq F_\alpha(p, n-p-1)$, $P = P(F \geq f)$.

Definition (ANOVA table for multiple regression)

Source	SS	d.f.	MS	F	P-value
Regression	SSR	p	MSR	$f = \text{MSR}/\text{MSE}$	$P(F \geq f)$
Residual	SSE	$n-p-1$	MSE		
Total	SST	$n-1$			

Example 8-11 (Significance of the long-jump model)

For Example 8-10, SST = 13503.6 and SSE = 2936.9. Test the regression function at level 5% and compute R^2.

Solution. SSR = SST − SSE = 10566.7; with $p = 2$ the d.f. are $9, 7, 2$. MSR = $10566.7/2 = 5283.35$, MSE = $2936.9/7 = 419.557$, $f = 12.593 \geq F_{0.05}(2,7) = 4.737$ ($P = 0.0048$): the regression function is significant.

Source	SS	d.f.	MS	F	P
Regression	10566.7	2	5283.35	12.593	0.0048
Residual	2936.9	7	419.557		
Total	13503.6	9			

$R^2 = \text{SSR}/\text{SST} = 0.7825$: about 78% of the variation is explained by the regression function.

Inference as in Example 8-11 also requires checking the model first, again with studentized residuals $\hat{e}_{st,i} = (y_i - \hat{y}_i)/\widehat{\text{SD}}(Y_i - \hat{Y}_i)$. Since there are several explanatory variables, the residual plot uses the fitted values on the horizontal axis: the points $(\hat{y}_i, \hat{e}_{st,i})$. For Example 8-10 this plot shows no problem, so the inference of Example 8-11 is trustworthy.